

LZ $0\nu\beta\beta$ Sensitivity: Key Parameters and Background Summary

Extracted from arXiv:1912.04248v2

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Abstract

This note summarizes the key detector assumptions, background model, and projected sensitivity of the LUX-ZEPLIN (LZ) experiment to neutrinoless double beta decay of ^{136}Xe , as reported in arXiv:1912.04248v2. LZ projects a median 90% CL exclusion sensitivity of 1.06×10^{26} years for the ^{136}Xe $0\nu\beta\beta$ half-life after 1000 live-days, with a total expected background of ~ 35.6 events in the $\pm 1\sigma$ ROI and inner 967 kg fiducial volume.

1 Detector Parameters

LZ is a two-phase liquid xenon TPC with 7 tonnes of active LXe, yielding 623 kg of ^{136}Xe at 8.9% natural abundance. The TPC drift region spans 145.6 cm (cathode to gate) with an inner diameter of 145.6 cm. Two PMT arrays (253 top, 241 bottom) detect scintillation (S1) and electroluminescence (S2) signals.

The detector is surrounded by an LXe skin veto (~ 2 tonnes, 100 keV threshold over $>95\%$ of the volume), an outer detector (Gd-doped liquid scintillator, 100 keV threshold, 1 μs coincidence window), and a water shield.

2 Key Assumptions

Table 1 collects the analysis assumptions that drive the sensitivity.

Table 1: Key analysis assumptions for the LZ $0\nu\beta\beta$ search.

Parameter	Value
$Q_{\beta\beta}$ (^{136}Xe)	2457.83 ± 0.37 keV
Energy resolution (σ/E at $Q_{\beta\beta}$)	1.0% (conservative)
Predicted resolution (NEST + LZ baseline)	0.88%
XENON1T demonstrated resolution	0.79%
ROI ($\pm 1\sigma$ around $Q_{\beta\beta}$)	$2433.3 < E < 2482.4$ keV
Extended energy range (PLR fit)	$2000 < E < 2700$ keV
Minimum vertex separation (Δz)	3 mm
Signal efficiency (after SS + veto cuts)	80%
Inner fiducial volume (cut-and-count)	967 kg ($26 < z < 96$ cm, $r < 39$ cm)
Extended fiducial volume (PLR)	5600 kg ($2 < z < 132.6$ cm, $r < 68.8$ cm)
Live time	1000 days
^{136}Xe exposure (natural abundance)	1360 kg-years
Skin veto threshold	100 keV
Outer detector veto threshold	100 keV
Veto coincidence window	1 μs

3 Background Model

3.1 Dominant gamma lines near $Q_{\beta\beta}$

Two gamma lines dominate the background near $Q_{\beta\beta}$:

- ^{208}Tl at 2614.5 keV (^{232}Th chain, BR = 35.9%): ~ 157 keV above $Q_{\beta\beta}$. Partially rejected by single-scatter cuts (the coincident ≥ 583 keV companion gamma tags multi-site events) and by skin/OD veto.
- ^{214}Bi at 2447.7 keV (^{238}U chain, BR = 1.5%): ~ 10 keV below $Q_{\beta\beta}$. Cannot be separated from the signal by energy resolution alone. The low branching ratio is fortunate.

3.2 Backgrounds in the $\pm 1\sigma$ ROI (1000 days, inner 967 kg)

Table 2 reproduces the complete background budget from Table I of the LZ $0\nu\beta\beta$ paper.

Table 2: Expected background counts in the $\pm 1\sigma$ ROI (2433–2482 keV) and inner 967 kg fiducial volume for 1000 live-days, assuming 1.0% energy resolution at $Q_{\beta\beta}$ and 3 mm single-scatter rejection in z . Activities marked † are upper limits.

Component	Mass (kg)	^{238}U -late (mBq/kg)	^{232}Th -late (mBq/kg)	Counts from ^{238}U	Counts from ^{232}Th	Total Counts
TPC PMTs	91.90	3.22	1.61	2.95	0.10	3.05
TPC PMT bases	2.80	75.90	33.10	1.52	0.03	1.55
TPC PMT structures	166.00	1.60	1.06	2.65	0.12	2.77
TPC PMT cables	88.70	4.31	0.82	1.44	0.19	1.63
Skin PMTs + bases	8.59	46.00	14.90	0.75	0.02	0.78
PTFE walls	184.00	0.04	0.01	0.39	0.00	0.39
TPC sensors	5.02	5.82	1.88	1.19	0.00	1.19
Field grids + holders	89.10	2.63	1.46	0.62	0.11	0.73
FC resistors	0.06	1350.00	2010.00	2.63	0.03	2.65
FC rings †	93.00	0.35	0.24	0.82	0.00	0.82
Ti cryostat †	2590.00	0.08	0.22	1.30	0.20	1.49
Cryo. insulation †	13.80	11.10	7.79	0.90	0.04	0.94
Outer detector †	22 900.00	4.71	3.73	1.70	1.08	2.79
Other components	438.00	1.83	1.65	2.10	0.31	2.41
Det. components subtotal				21.0	2.32	23.3
Cavern walls	29 000.00	3.21	8.41	—	—	11.60
^{137}Xe (neutron-induced)						0.28
Internal ^{222}Rn						0.45
^{136}Xe $2\nu\beta\beta$						0.01
^8B solar ν						0.03
Total				24.2	10.7	35.6

3.3 Detailed background rates per year

Table 3 converts the 1000-day counts from Table 2 to annual rates per component and per isotope chain (scaling factor $365.25/1000 = 0.36525$).

Key observations from the annual rates:

Table 3: Annual background rates (counts/year) in the $\pm 1\sigma$ ROI and inner 967 kg volume, broken down by component and decay chain. Derived from Table 2.

Component	$^{238}\text{U-late}$ (cts/yr)	$^{232}\text{Th-late}$ (cts/yr)	Total (cts/yr)
TPC PMTs	1.077	0.037	1.114
TPC PMT bases	0.555	0.011	0.566
TPC PMT structures	0.968	0.044	1.012
TPC PMT cables	0.526	0.069	0.596
Skin PMTs + bases	0.274	0.007	0.285
PTFE walls	0.142	0.000	0.142
TPC sensors	0.435	0.000	0.435
Field grids + holders	0.226	0.040	0.267
FC resistors	0.961	0.011	0.968
FC rings	0.300	0.000	0.300
Ti cryostat	0.475	0.073	0.544
Cryo. insulation	0.329	0.015	0.343
Outer detector	0.621	0.395	1.019
Other components	0.767	0.113	0.881
Det. components subtotal	7.67	0.85	8.51
Cavern walls ($^{238}\text{U} + ^{232}\text{Th}$)	1.17	3.06	4.24
External subtotal	8.84	3.91	12.75
^{137}Xe (neutron-induced)	—	—	0.102
Internal ^{222}Rn	—	—	0.164
^{136}Xe $2\nu\beta\beta$	—	—	0.004
^8B solar ν	—	—	0.011
Internal subtotal	—	—	0.28
Grand total	8.84	3.91	13.01

- $^{238}\text{U-late}$ dominates over $^{232}\text{Th-late}$ by a factor of ~ 9 for detector components, because the ^{214}Bi 2448 keV line falls directly under $Q_{\beta\beta}$ while the ^{208}Tl 2615 keV line is partially rejected by veto and single-scatter cuts.
- For cavern walls the ratio reverses: ^{232}Th contributes $\sim 2.6\times$ more than ^{238}U , reflecting both the higher ^{232}Th activity in the rock (8.41 vs. 3.21 Bq/kg) and the penetrating power of the 2615 keV line through the shielding.
- The top three detector-component contributors per year are: TPC PMTs (1.11), outer detector (1.02), and PMT structures (1.01), followed closely by FC resistors (0.97).
- Internal backgrounds ($^{222}\text{Rn} + ^{137}\text{Xe} + 2\nu\beta\beta + ^8\text{B}$) contribute only 0.28 counts/year ($\sim 2\%$ of total), making the search overwhelmingly externally-background-limited.

3.4 Background composition

The dominant contributions to the total 35.6 counts in 1000 days are:

1. **Detector components** (23.3 counts, 65%): dominated by TPC PMTs (3.05), PMT structures (2.77), FC resistors (2.65), other components (2.41), and PMT cables (1.63). The $^{238}\text{U-late}$ chain contributes $\sim 9\times$ more than $^{232}\text{Th-late}$ from detector components,

because the ^{214}Bi line at 2448 keV sits directly under $Q_{\beta\beta}$ while the ^{208}Tl line at 2615 keV is partially rejected by single-scatter and veto cuts.

2. **Cavern walls** (11.6 counts, 33%): the rock gamma flux is the single largest individual source. Additional steel shielding (8 cm above the water tank) was added specifically for the $0\nu\beta\beta$ program.
3. **Internal backgrounds** (0.77 counts, 2%): ^{222}Rn (0.45), ^{137}Xe from neutron capture (0.28), $2\nu\beta\beta$ (0.01), ^8B solar neutrinos (0.03).

4 Sensitivity

Using a profile likelihood ratio analysis over the extended fiducial volume (5.6 tonnes, 1360 kg.years of ^{136}Xe exposure) and the energy range 2000–2700 keV:

- **Natural abundance** (8.9% ^{136}Xe): median 90% CL exclusion sensitivity $T_{1/2}^{0\nu} > 1.06 \times 10^{26}$ years after 1000 live-days.
- **90% enrichment**: $T_{1/2}^{0\nu} > 1.06 \times 10^{27}$ years after 1000 live-days (same detector, no other upgrades).
- **Effective neutrino mass**: $m_{\beta\beta} = 53\text{--}164$ meV (range from nuclear matrix element uncertainty).

4.1 Sensitivity dependence on key parameters

- **Energy resolution**: modest impact from 1.0% to $\sim 1.4\%$; significant degradation above 2.0% (the ^{208}Tl peak leaks into the ROI).
- **Vertex separation Δz** : strong dependence. Single-scatter rejection is the most powerful background cut for external gammas at these energies. Below ~ 2 mm, signal efficiency begins to drop (bremsstrahlung from the two electrons creates resolvable multi-site topologies).
- **PLR vs. cut-and-count**: the profile likelihood analysis with extended volume and energy range improves sensitivity by a factor of ~ 2 over a simple cut-and-count in the inner 967 kg.

5 Key Numbers for LXeMC Simulation

The following parameters from this analysis are used directly in the LXeMC simulation:

Table 4: Parameters adopted by LXeMC from the LZ $0\nu\beta\beta$ analysis.

Parameter	Value	Use in LXeMC
FV dimensions	$r < 39$ cm, $26 < z < 96$ cm	FVGeometry
Energy resolution	$\sigma/E = 1.0\%$ at $Q_{\beta\beta}$	event smearing
Δz clustering	3 mm	SS/MS classification
Skin veto threshold	100 keV	veto_skin
TPC veto threshold	10 keV	veto_TPC
ROI	$\pm 1\sigma$ around $Q_{\beta\beta}$	event selection
Component masses	Table 2	source geometry JSON
Component activities	Table 2	source flux generation